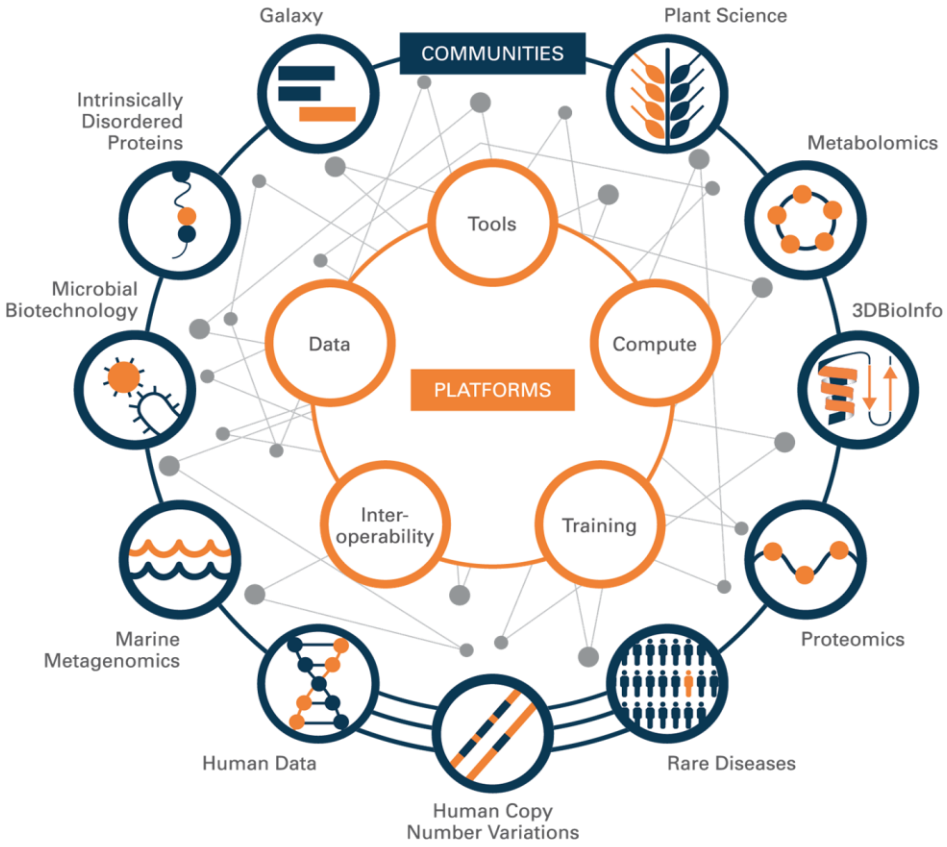
Do you need help with life science data?

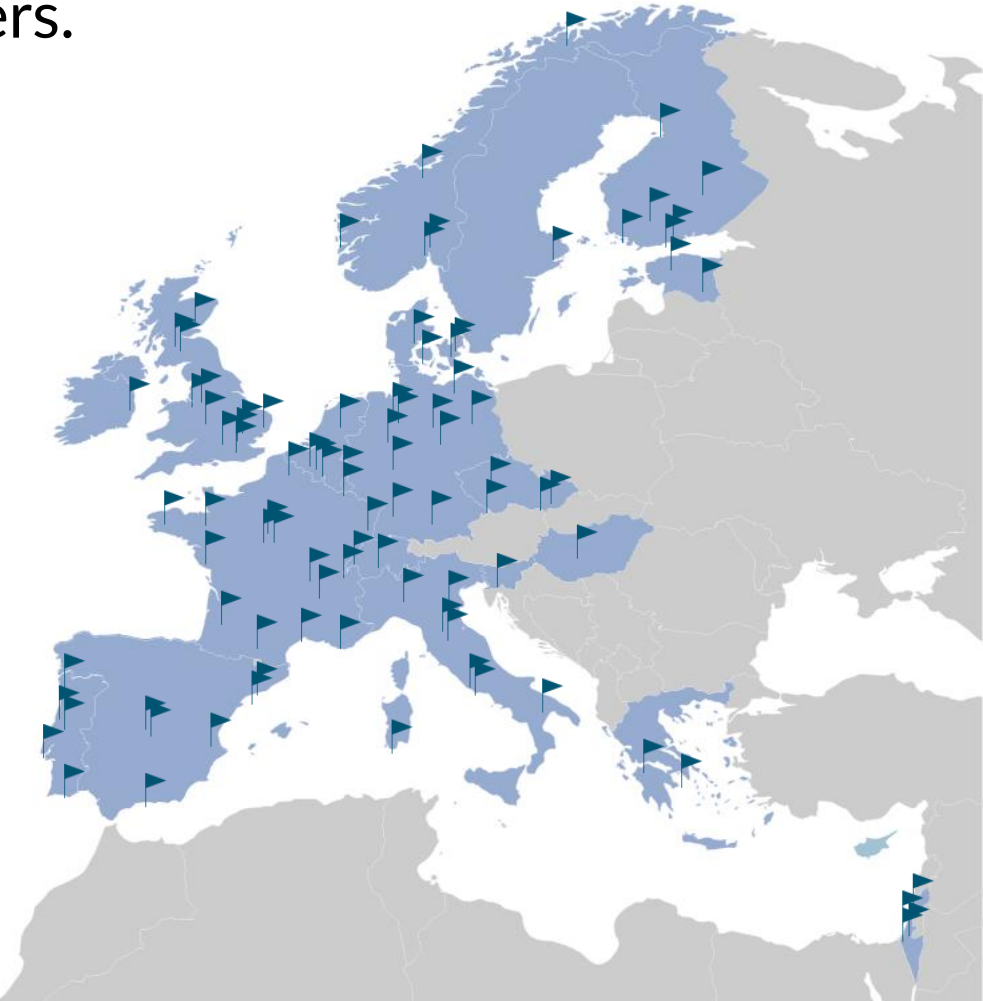
Get it by ELIXIR-Norway

Arturo Vera Ponce de Leon
ELIXIR Norway
Centre for Integrative Genetics (CIGENE)
Norwegian University of Life Sciences

ELIXIR - A distributed infrastructure for Life Science data

ELIXIR is an intergovernmental organization that brings together life science resources from across Europe. The resources include databases, software tools, training materials, cloud storage and supercomputers.



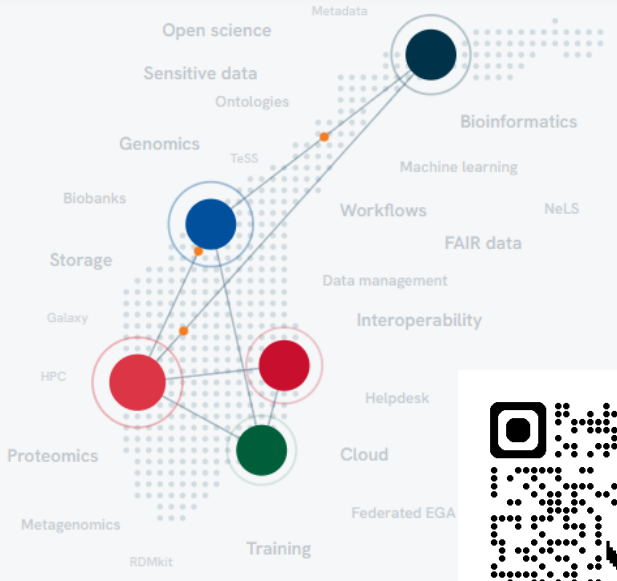


Slide by David Dolan

ELIXIR Norway

About | Research Support | Services | Events | Training | Funding & Projects | News

Collaborative Life Science Hub Linking Five Organisations to ELIXIR Europe



Coordinated from the University of Bergen, ELIXIR Norway connects the University of Oslo, UiT Tromsø, NTNU, and NMBU into a national network — part of ELIXIR Europe's infrastructure for life science data.

Organisation

ELIXIR Norway is organised across five partner universities, with dedicated leadership, coordination, and advisory committees guiding our national infrastructure.

Bergen | Oslo | Tromsø | Ås | Trondheim

https://elixir.no/

HELPDESK



Contact our nationally coordinated Helpdesk to get advise and support on most topics regarding bioinformatics and data management.

How to get help?

Support model

- Projects that can be solved within a (two) working day are **free of charge!!**

Contact us by email at support@elixir.no

Examples of topics ELIXIR-NMBU can help

NB! ELIXIR-Norway is not University-IT

- Assistance analyzing ‘omics data (genomics, transcriptomics, proteomics and metabolomics)
- Assistance with scripting and coding (R, Python, Bash, Perl, etc.)
- Assistance submitting ‘omics data to public repositories (ENA, GenBank, PRIDE)

Topics ELIXIR-NMBU can help with.

- Assistance using Linux/Unix servers and high-performance computing (HPC) environments (Orion or Sigma2)
- Allocation of storage space (up to 10 Tb) at the Norwegian e-Infrastructure for Life Sciences (NeLS)
- Assistance applying for resources at National e-infrastructure services (Sigma2) for HPC usage and storage

Topics ELIXIR-NMBU can help with.

- Assistance using the platform Galaxy for bioinformatics
- Assistance creating and developing data-management plan (DMP)
- Providing courses on bioinformatics and omics data analysis

ELIXIR-NMBU **cannot** help with

NB! ELIXIR-Norway is not IT

- Installing software in the Sigma2 servers
- Provide HPC computational power
- Copy or move data between Orion and Sigma2 servers
- Helping with the printer when is not working



What if do I need assistance for more than 2 days?

- **Projects that require an effort beyond technical assistance/consultancy may be set up as a research collaboration (e.g. by providing funding for a percentage of the position of a helpdesk member).**



ELIXIR-How to get help?

If your research group is generating life science data using molecular analyses and 'omics technologies, we exist to support your current and future efforts in data management, data storage, and data analyses.

Contact us by email at support@elixir.no

Scan me

HELPDESK

Contact our nationally coordinated Helpdesk to get advise and support on most topics regarding bioinformatics and data management.

Where else can ELIXIR Norway help?



IDEA
APPLICATION

PUBLICATION(S)





DATA



DATA STORAGE

NeLS
Norwegian e-Infrastructure for Life Sciences



EXPERIMENT



METADATA



DATA ANALYSIS



DATA DEPOSITION

DATA MANAGEMENT PLAN



DSW

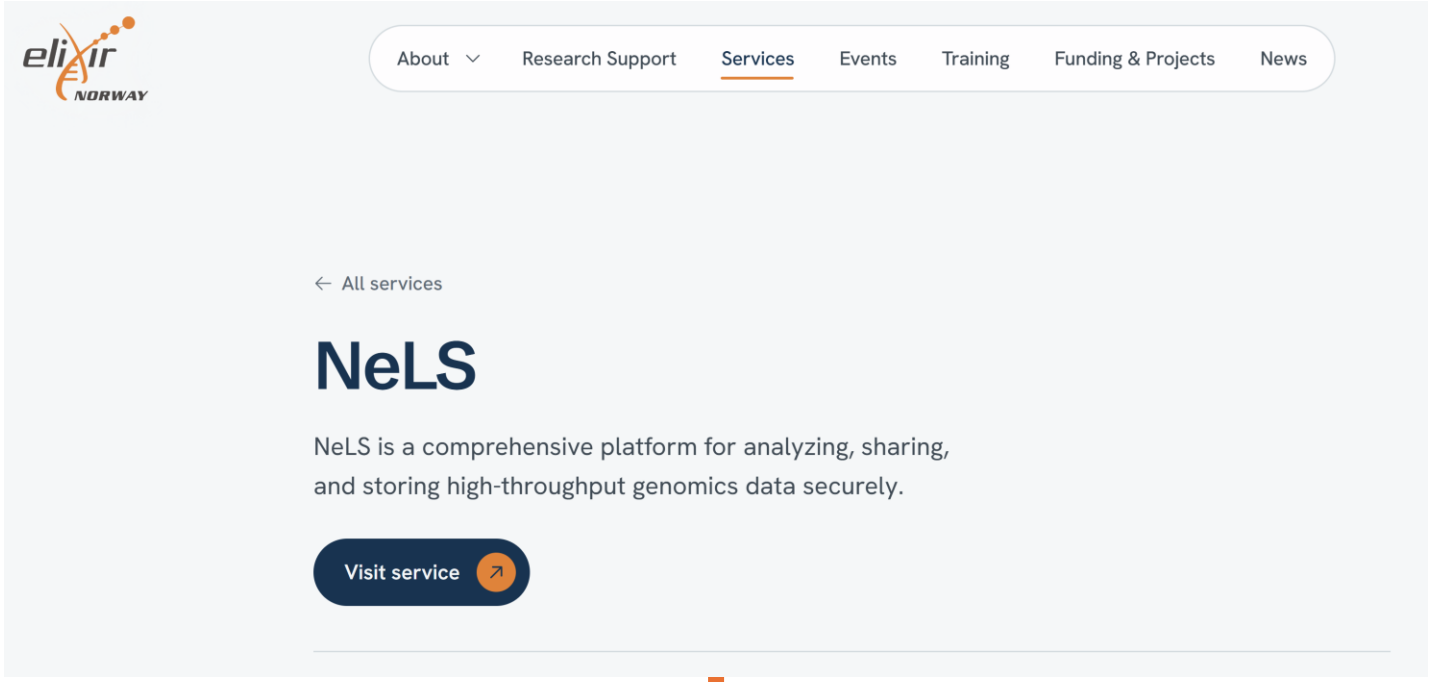
Do you need to storage/share large amount of data?

Get up to10Tb space in NeLS

Arturo Vera Ponce de Leon
ELIXIR Norway
Centre for Integrative Genetics (CIGENE)
Norwegian University of Life Sciences

NeLS by elixir NORWAY

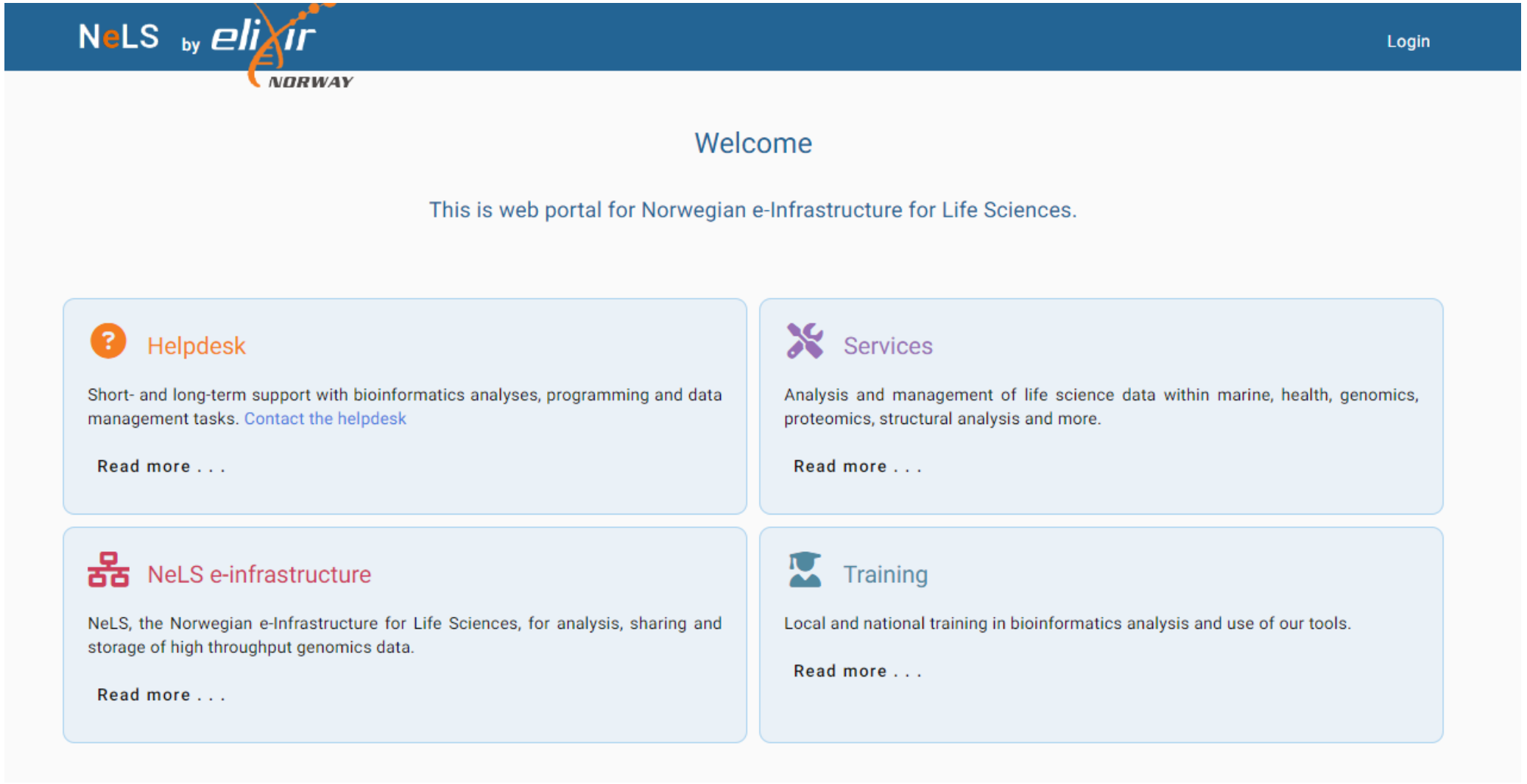




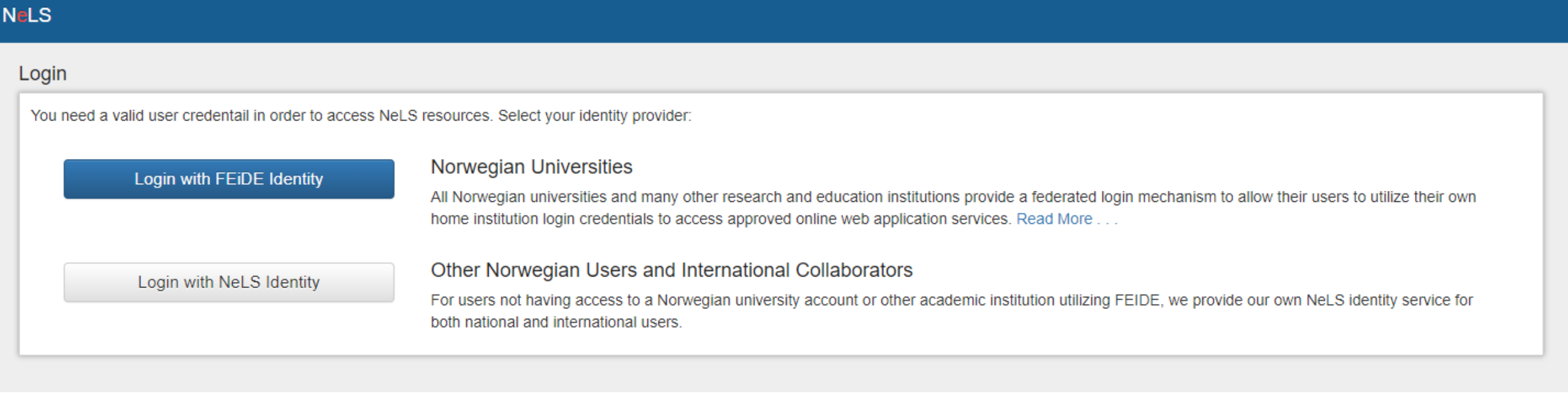
The Norwegian e-infrastructure for life sciences (NeLS), is a unique national portal for sharing, analysis and **midterm** storage (up to 10Tb) of **non-sensitive** molecular life science data.

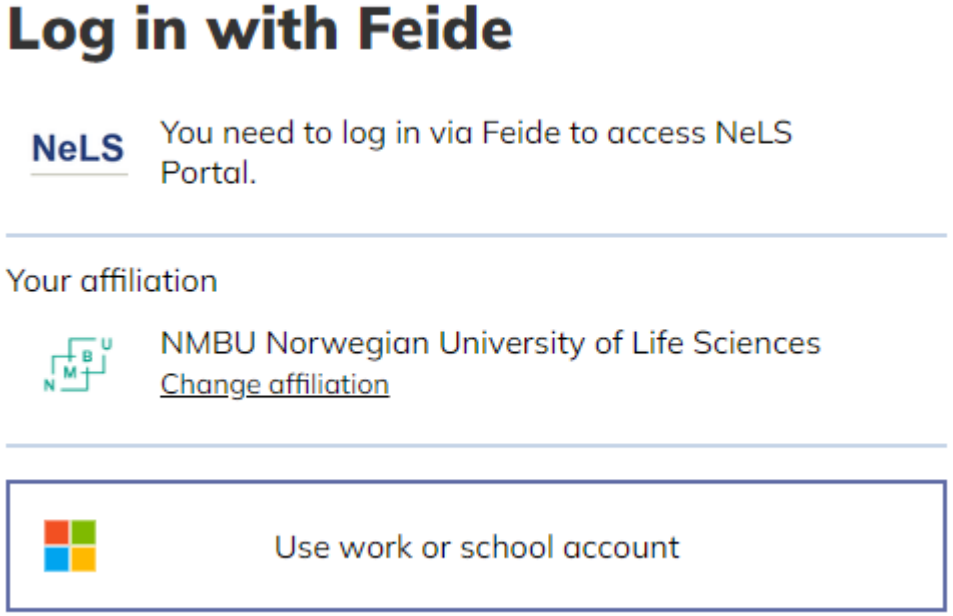
Login to NeLS using the WebPortal

https://nels.ELIXIR.no/nels-web/#/



Scan me!

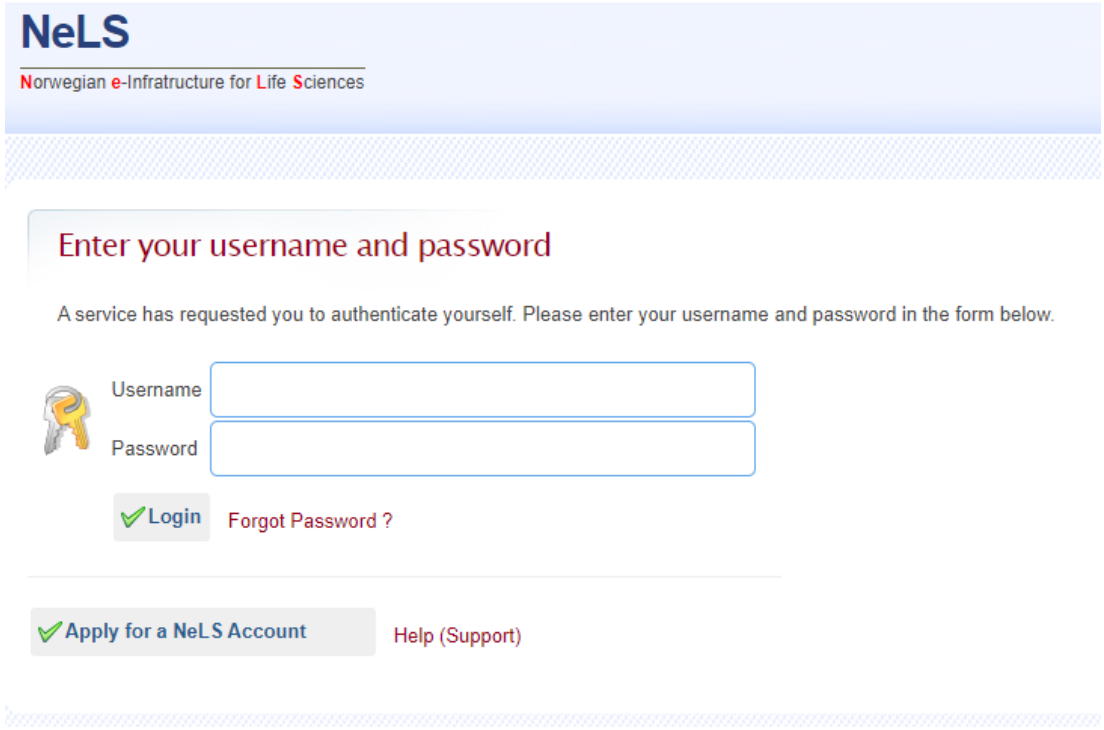


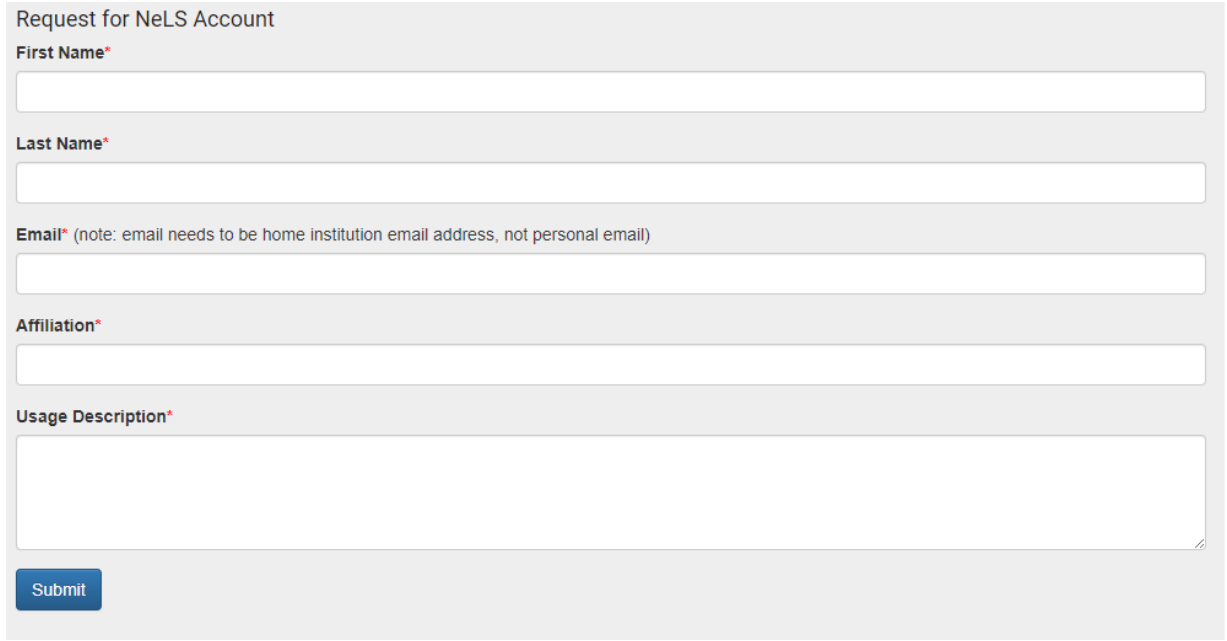


All users with FEIDE login credentials can access the NeLS Portal

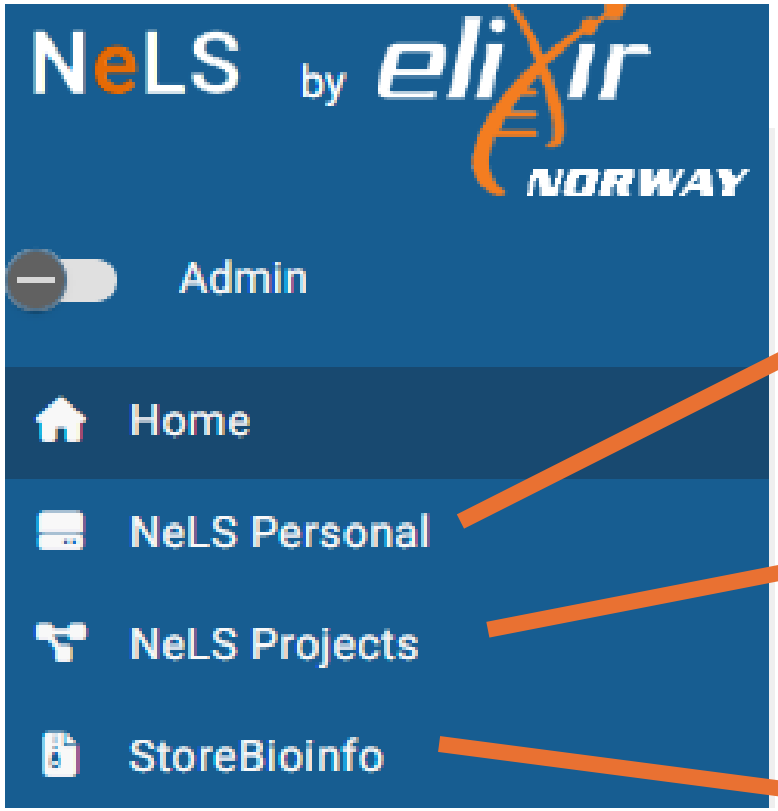
I don't have FEIDE login credentials (Non-Norwegian collaborators)

- It is also possible to login to NeLS via a NeLS idp. You can apply for a NeLS idp





The NeLS filesystem



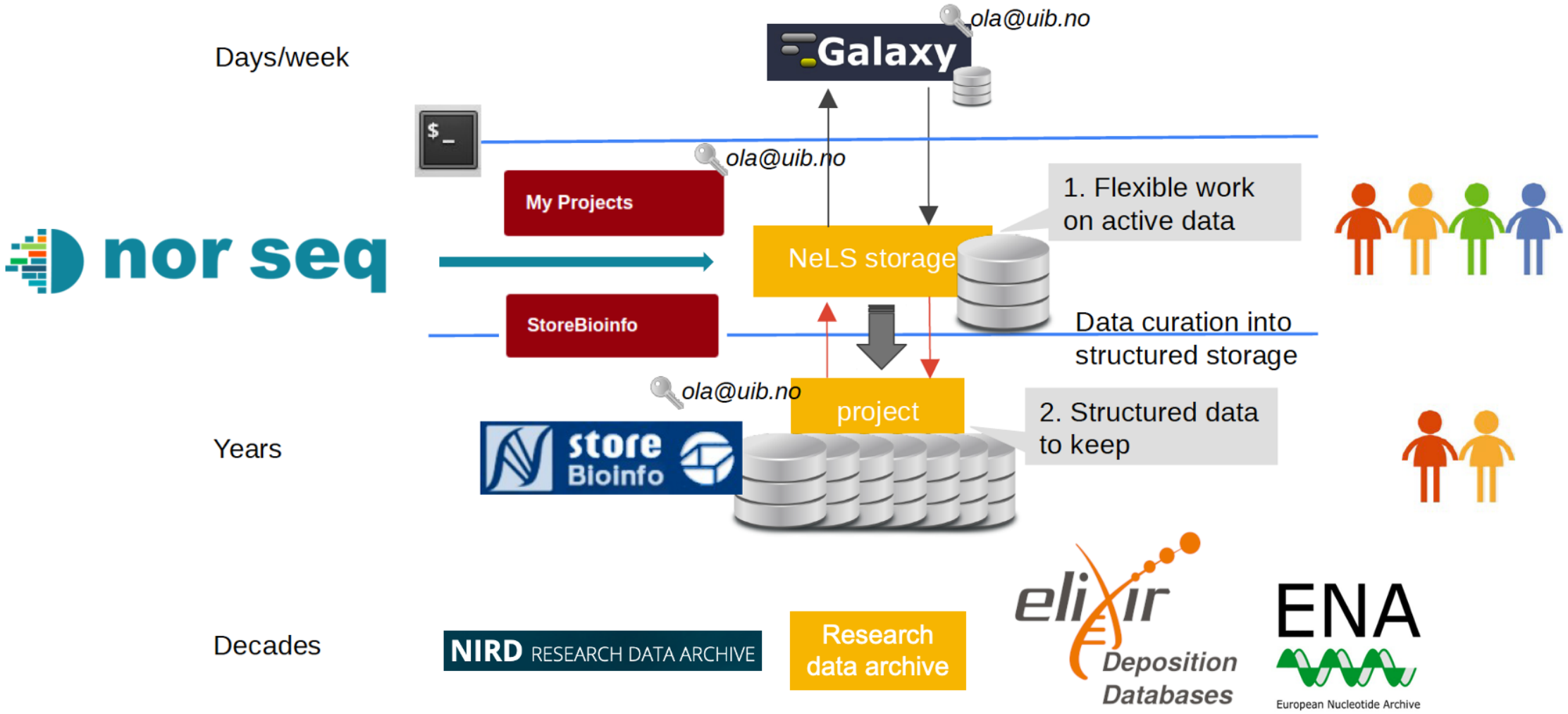
Personal area (from 1-10 Tb)
space **Not allowed to share**

Share area (1-10Tb) with multiple sub-folder
Sharing capability

StoreBioInfo (SBI) is for long term storage of project data.

NeLS
Norwegian e-Infrastructure for Life Sciences

How does NeLS work?



Apply for a project in NeLS for sharing data within and outside Norway

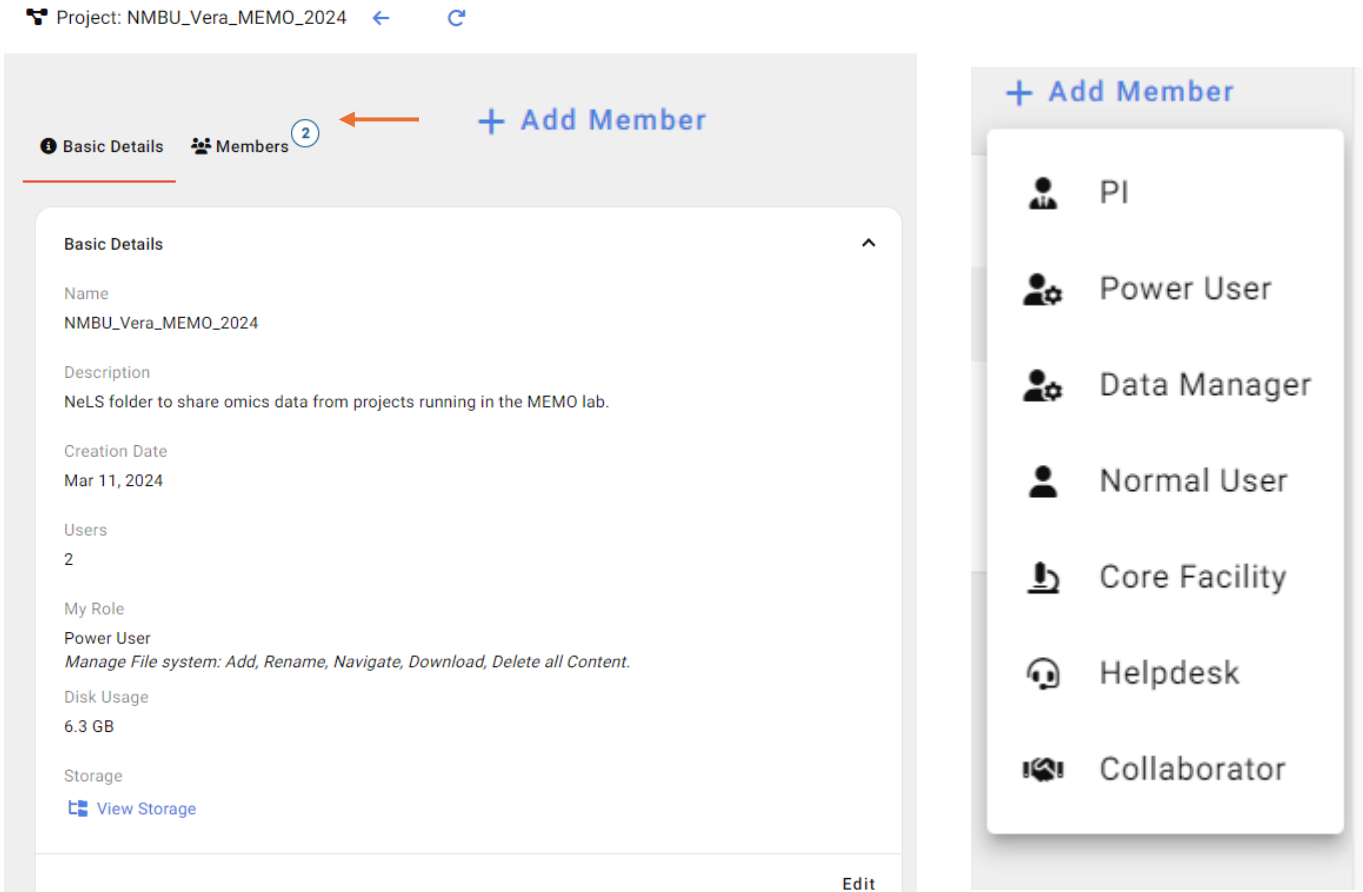
Send an email to support@elixir.no

You will get a form to fill, after fill it a new project will be created

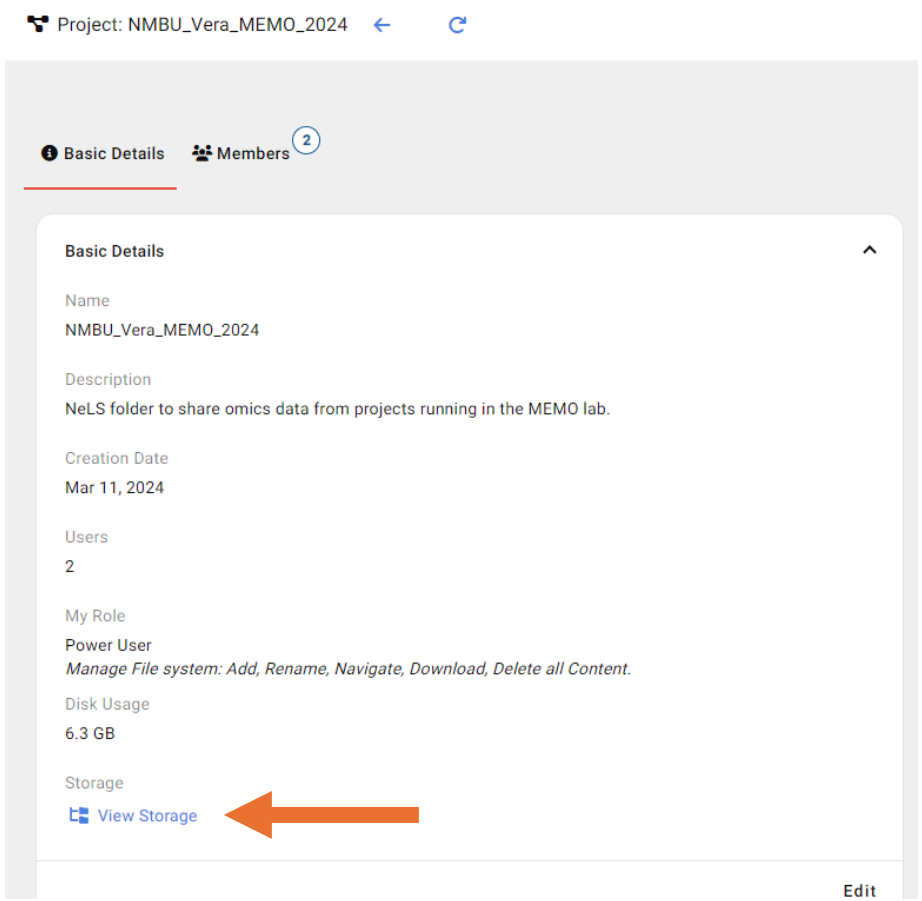
Adding members to a project

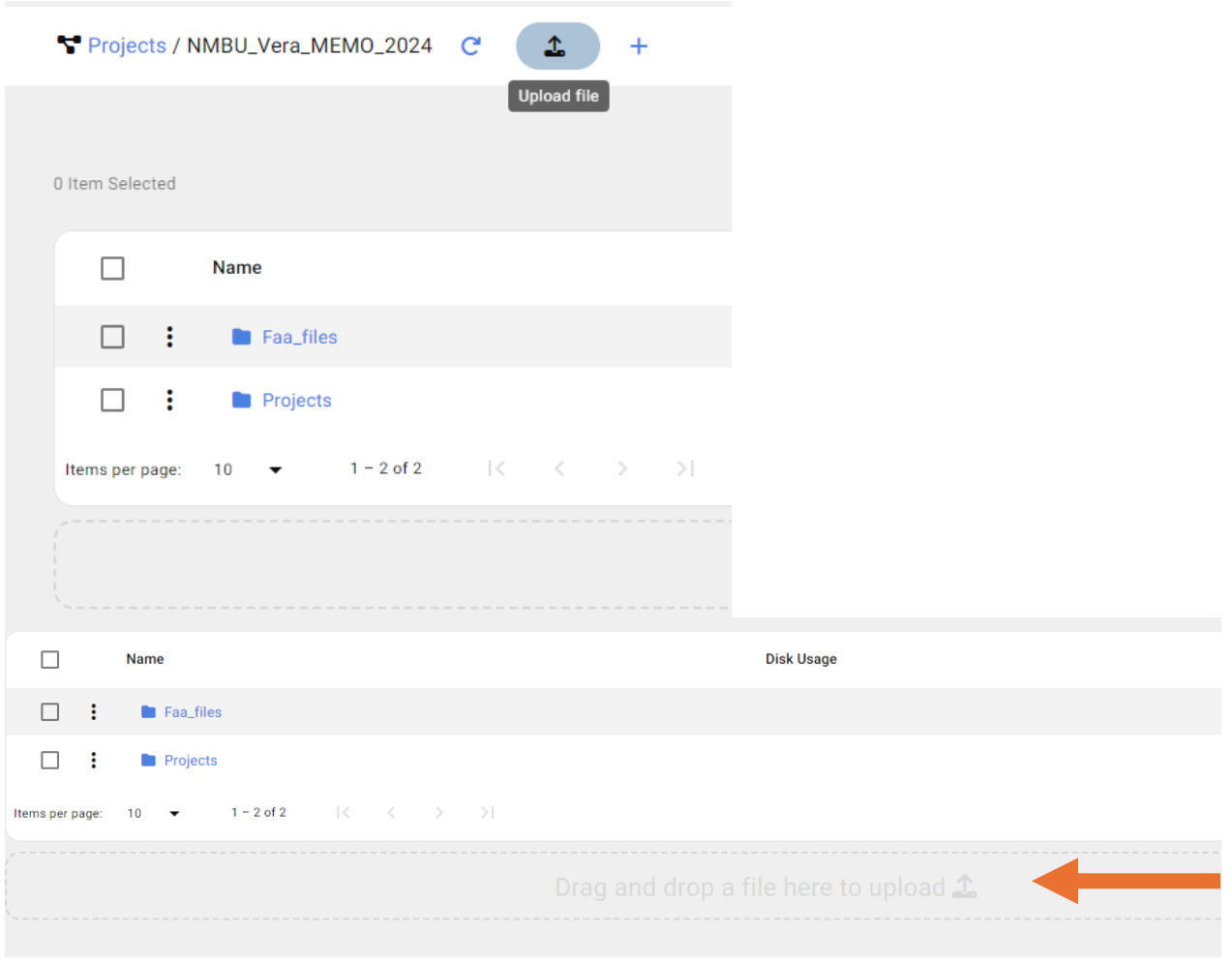
- If you have created a project in NeLS you can add members to that to share data.

- To add people, users must be register in NeLs before adding it to the project



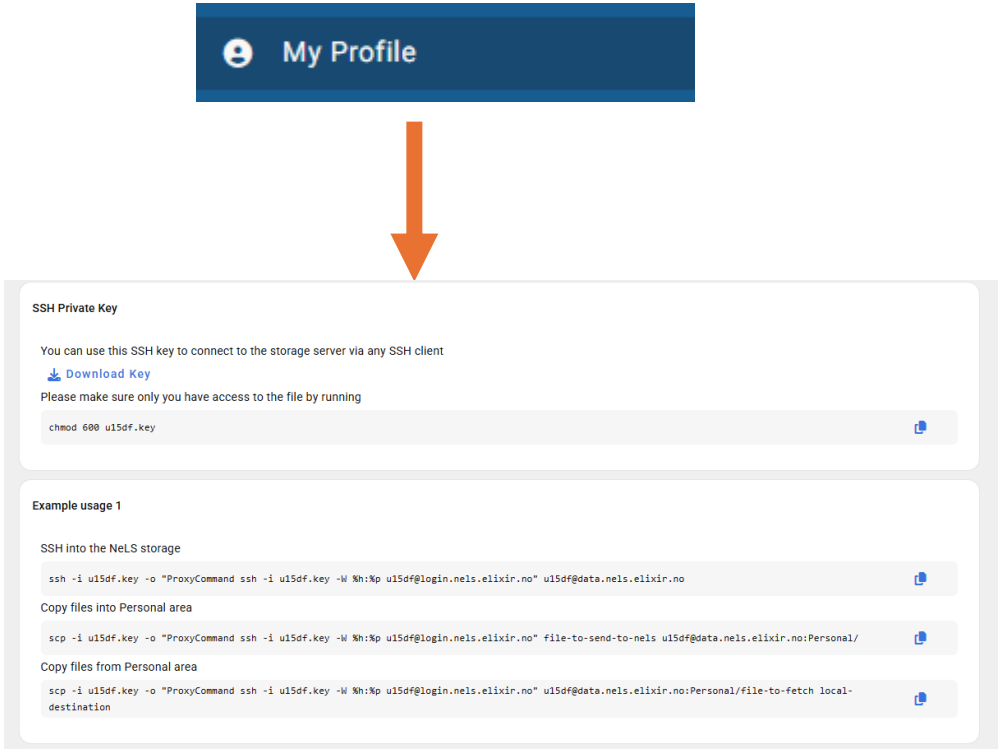
Transfer data to and from NeLS (Using the web-portal)





Using NeLS trough the CLI

- NeLS servers can be reached directly from the CLI using Secure Shell (SSH) protocol:



Using NeLS trough the CLI

```
(base) auve@L004538:NeLS$ ssh -i u15df@newstor.cbu.uib.no.key u15df@newstor.cbu.uib.no
Last login: Tue May 28 14:12:43 2024 from 128.39.239.131
FreeBSD 14.0-RELEASE-p6 (GENERIC) #0: Tue Mar 26 20:26:20 UTC 2024

Welcome to FreeBSD!

Release Notes, Errata: https://www.FreeBSD.org/releases/
Security Advisories:   https://www.FreeBSD.org/security/
FreeBSD Handbook:      https://www.FreeBSD.org/handbook/
FreeBSD FAQ:           https://www.FreeBSD.org/faq/
Questions List:        https://www.FreeBSD.org/lists/questions/
FreeBSD Forums:        https://forums.FreeBSD.org/

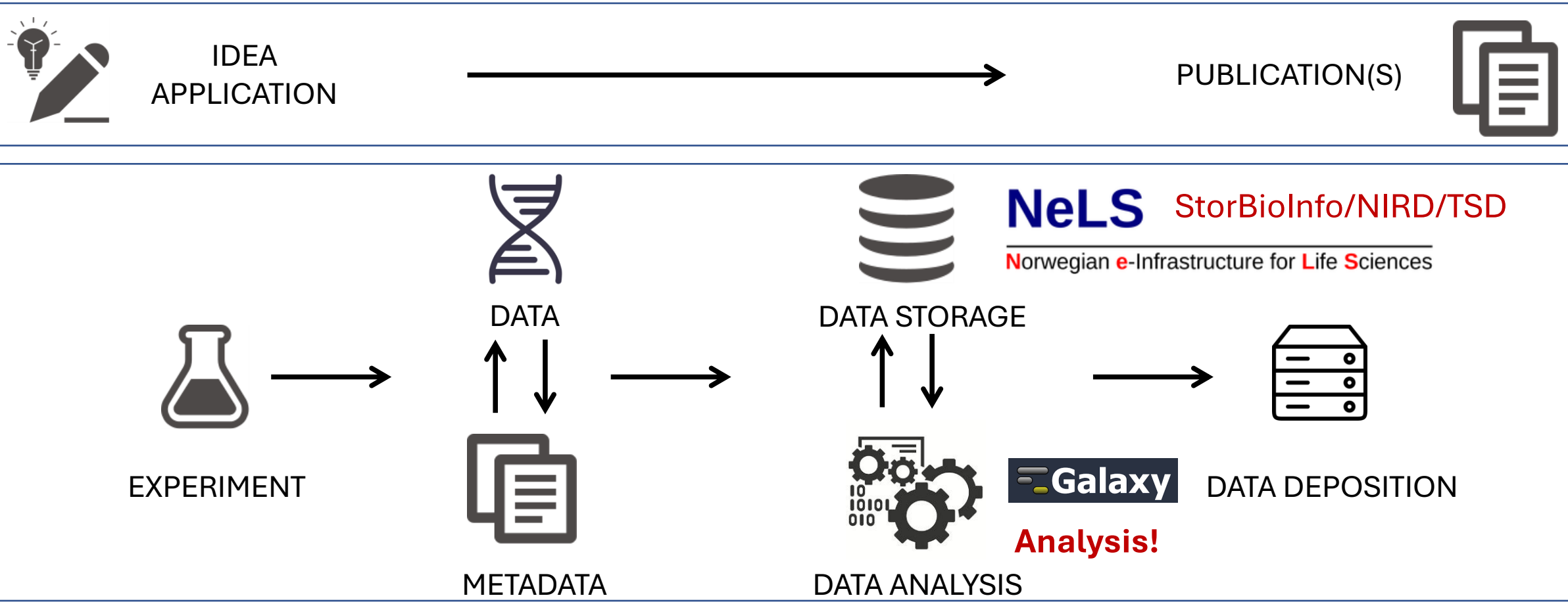
Documents installed with the system are in the /usr/local/share/doc/freebsd/
directory, or can be installed later with:  pkg install en-freebsd-doc
For other languages, replace "en" with a language code like de or fr.

Show the version of FreeBSD installed:  freebsd-version ; uname -a
Please include that output and any error messages when posting questions.
Introduction to manual pages:  man man
FreeBSD directory layout:      man hier

To change this login announcement, see motd(5).
[u15df@newstor01: ~]$
```

usegalaxy.no

elixir NORWAY


IDEA
APPLICATION
PUBLICATION(S)
NeLS
StorBioInfo/NIRD/TSD
Norwegian e-Infrastructure for Life Sciences
DATA
DATA STORAGE
EXPERIMENT
METADATA
DATA ANALYSIS
Galaxy
Analysis!
DATA DEPOSITION
DATA MANAGEMENT PLAN
DSW

usegalaxy.no – The national Galaxy server

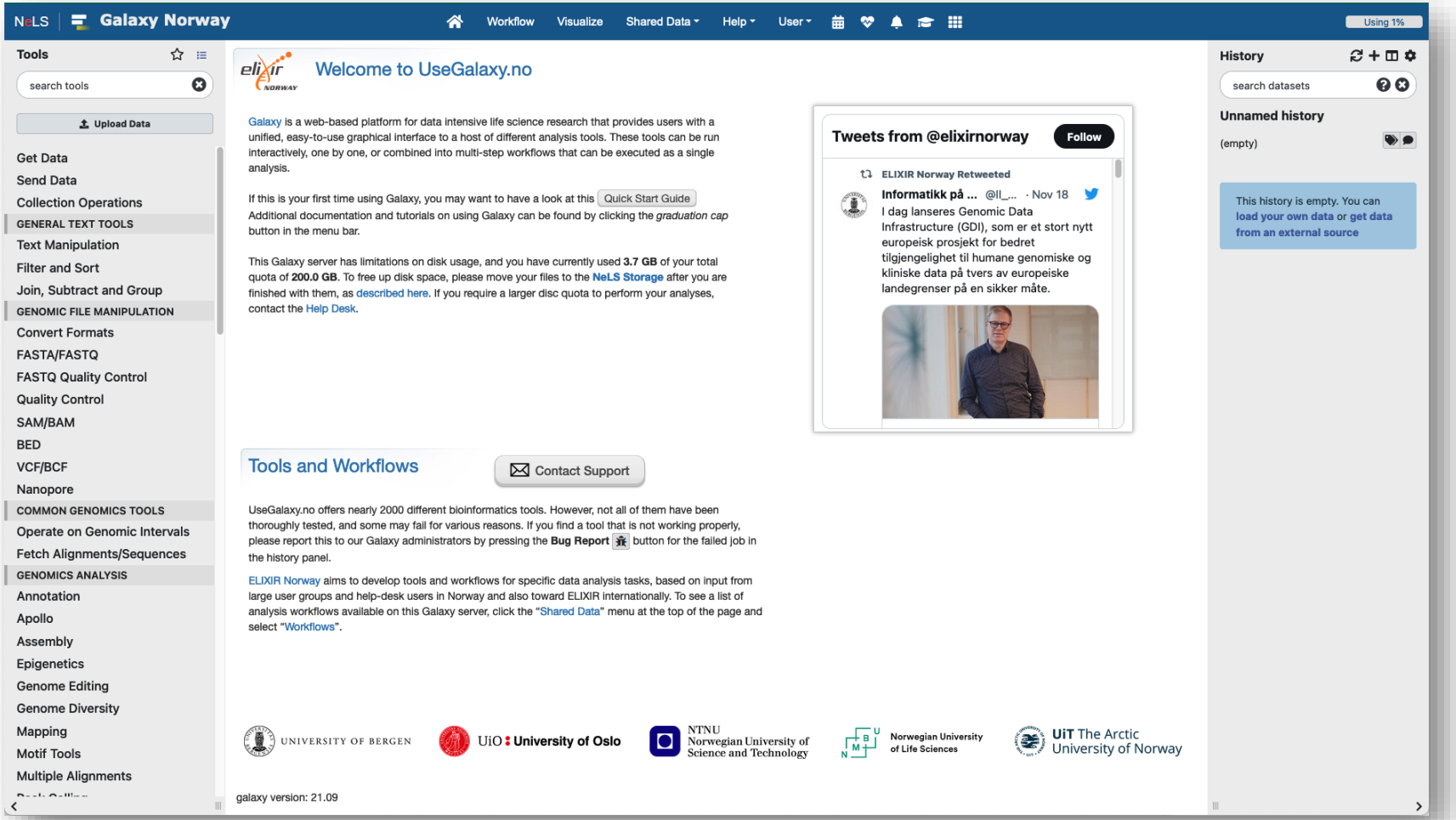
Web-based platform

Provide bioinformatic tools and workflows

Open to all Norwegian users and collaborators

Enable accessible, reproducible, and transparent computational biomedical research

Directly connected to the NeLS storage



usegalaxy.no – The basics

Important features:

Tools menu with ~2000 tools sorted in sections

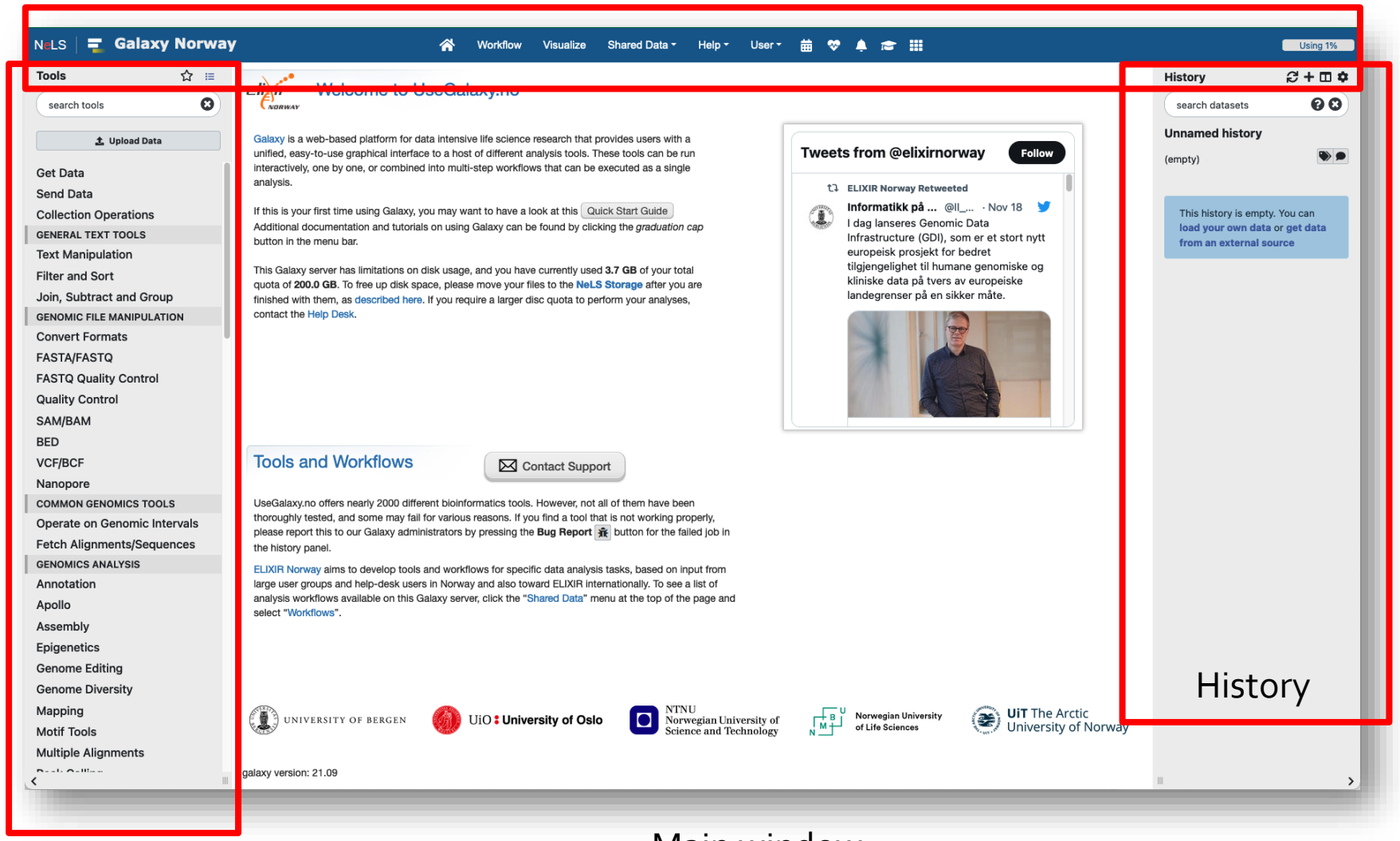
Current disk usage (default is 200 GB total personal disk space)

Status of compute backend

Quick start guide

Contact support

Help and guides

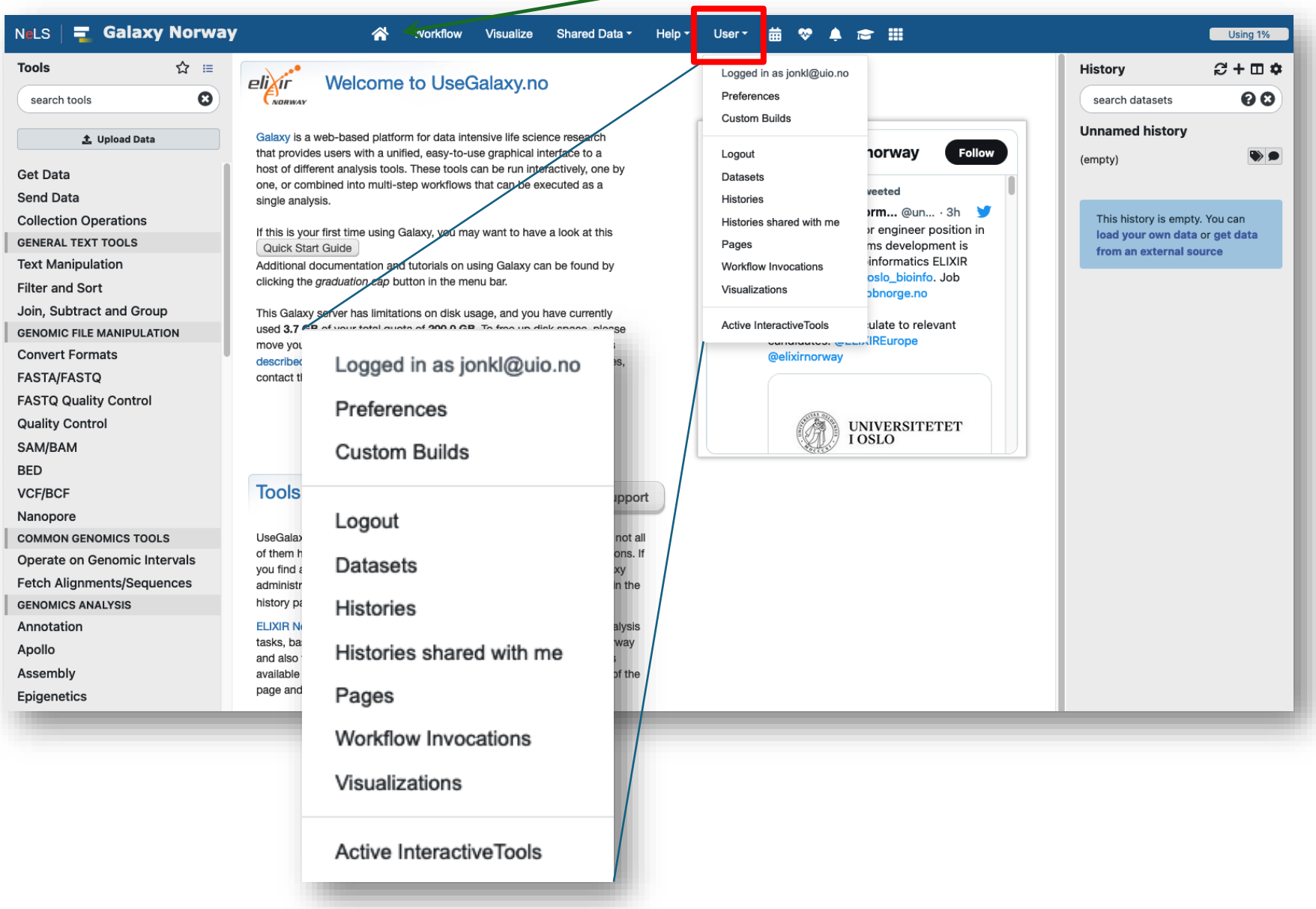


Your account and saved data

Account settings and saved data such as saved histories and visualizations

Histories other users have shared with you

Log of analysis workflows you have run



Hands on Galaxy

How to perform RNAseq analysis and Differential Gene Expression in Galaxy.no

https://training.galaxyproject.org/training-material/topics/transcriptomics/tutorials/ref-based/tutorial.html